

KALAPANENI RITESH CHOWDARY

AI & ML Engineering Student | Computer Vision & GenAI Developer

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SUMMARY

AI & ML engineering student with hands-on experience in computer vision, Generative AI, and web-based AI applications. Experienced with OpenCV, YOLOv8, ALPR, Gemini API, Flask, and prompt engineering, with projects focused on traffic safety, healthcare, and education. 4× Oracle-certified in Generative AI, Agentic AI, APEX, and SQL.

EDUCATION

R.M.D Engineering College | B.Tech in Artificial Intelligence & Machine Learning | 2024–2028 | CGPA: 7.22/10

Narayana Junior College | Intermediate (MPC) | 2022–2024 | 86%

EXPERIENCE

Computer Vision & AI Developer Intern | Doneshswari Technologies

2025 – Present | Tirupati, India

- Developed OpenCV and YOLOv8-based real-time video processing pipelines for traffic signal recognition and vehicle detection.
- Implemented computer vision modules for automatic license plate recognition (ALPR) and traffic violation detection, including helmetless riding and mobile-phone usage identification.
- Integrated vision-processing modules with web-based analytics interfaces for monitoring detected events.
- Applied image preprocessing and object-tracking techniques to improve real-time detection performance.

AI Developer Intern (Project-Based) | AI & ML Engineering Initiatives

2025

- Developed an AI Career Counselor web application using Python, Flask, JavaScript, and Google Gemini API.
- Designed structured prompts to generate personalized career guidance based on user inputs.
- Built responsive interfaces for collecting user information and displaying AI-generated recommendations.

Frontend Developer (Project Role) | Web Engineering Initiatives

2024 – 2025

- Developed responsive web interfaces using HTML, CSS, and JavaScript.
- Built reusable frontend components and improved UI/UX usability.
- Optimized page responsiveness and frontend performance across devices.

PROJECTS

Trivia-X — AI Traffic Violation Detection

Tech: Python, OpenCV, YOLOv8, ALPR, Computer Vision

- Developed a computer vision system for real-time traffic video analysis using OpenCV and YOLOv8.
- Implemented vehicle detection, traffic violation identification, and automatic license plate recognition.
- Explored detection of helmetless riding and mobile-phone usage from traffic footage.

AI Career Counselor

Tech: Python, Flask, Gemini API, JavaScript, Prompt Engineering

- Developed an AI-powered career guidance application using Flask and Google Gemini API.
- Implemented structured prompt engineering to generate personalized career recommendations.
- Built a responsive web interface for user input and dynamic AI-generated guidance.

Hospital Bed & Emergency Management System

Tech: Python, Flask, JavaScript, AI, GPS, HTML/CSS

- Designed a multi-portal healthcare platform for citizens, doctors, ambulances, and government officials.
- Developed system architecture around hospital bed availability and emergency coordination.
- Incorporated concepts for AI-based emergency surge prediction and GPS-assisted ambulance routing.

Digital Guidance Platform for Government College Enrollment — SIH 2025

Tech: Python, Recommendation Systems, Web Architecture, UI/UX

- Developed a Smart India Hackathon 2025 project focused on helping students discover suitable government colleges.
- Designed recommendation logic based on academic eligibility, cutoff information, and career preferences.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, SQL

AI/ML: OpenCV, YOLOv8, Computer Vision, ALPR, KNN, Decision Trees, K-Means, Data Preprocessing, Model Evaluation

Generative AI: Google Gemini API, Generative AI, Prompt Engineering, Agentic AI

Web/Backend: Flask, REST APIs, HTML5, CSS3, JavaScript, Oracle APEX

Databases/Tools: Oracle SQL, PL/SQL, Oracle Cloud Infrastructure, Git, GitHub, VS Code

Fundamentals: Data Structures & Algorithms, Problem Solving, UI/UX, Debugging

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — Oracle University
- Oracle APEX Cloud Developer Certified Professional — Oracle University
- Agentic AI Certified Foundations Associate — Oracle University
- Oracle AI Database SQL Certified Associate — Oracle University